Law and Finance Matter: Lessons from Externally Imposed Courts

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This paper provides novel evidence on the real and financial market effects of legal institutions. Our analysis exploits persistent and externally imposed differences in court enforcement that arose when the U.S. Congress assigned state courts to adjudicate contracts on a subset of Native American reservations. Using area-specific data on small business lending, we find that reservations assigned to state courts, which enforce contracts more predictably than tribal courts, have stronger credit markets. Moreover, the law-driven component of credit market development is associated with significantly higher per capita income, with stronger effects in sectors that depend more on external financing. (*JEL* G21, K40, P48)

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How legal institutions relate to financial development and long-run growth is of central importance in economics. Nonetheless, the real and financial effects of legal systems continue to be widely debated (Levine 2005; Zingales 2015).

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Evaluating these effects is a significant challenge because numerous factors lead to cross-national differences in economic development (Sala-i-Martin, Doppelhofer, and Miller 2004; Dippel 2014). Moreover, institutions take shape alongside real outcomes, making it difficult to identify the causal linkages between law, finance, and growth.

Motivated by these concerns, we evaluate the long-run consequences of legal institutions using quasi-experimental variation in court enforcement in a within-country setting—Native American reservations in the United States. Evidence from this setting shows that stronger and more predictable contract enforcement leads to stronger credit markets, which in turn facilitates economic development. These differences matter. Economic development is lower on reservations than in areas nearby, but differences in court enforcement explain up to 70% of the reservation income gap.

Reservation economies are an ideal setting for studying the real and financial effects of legal institutions. Specifically, the U.S. Congress imposed sharp, long-run differences in court enforcement across reservations by passing Public Law 280 (PL280) in 1953. Congress proposed PL280 for reasons unrelated to credit markets and development prospects, but a provision added to the final version of the law assigned state courts to adjudicate contract disputes on a subset of reservations without consent from tribes (Anderson and Parker 2008). Meanwhile, reservations unaffected by PL280 settle disputes in their own tribal courts. In comparison to tribal courts, state courts provide stronger and more predictable contract enforcement, in part because their precedent is better understood (Mudd 1972; Parker 2012). Moreover, reservations exhibit substantially less heterogeneity in culture, geography, and trade than the cross-national setting. Because affected and unaffected reservations had similar credit market conditions prior to the law, the variation in adjudication arising from PL280 is a unique opportunity to test how legal enforcement affects credit markets and real economic activity.

The first stage of our empirical analysis shows that PL280 created long-lasting differences in credit market activity. Data on small business lending from the Federal Financial Institutions Examination Council (FFIEC) allow us to construct reservation-specific measures of business credit. On average, counties hosting a reservation that falls under state court jurisdiction have almost 40% more small business lending compared with corresponding counties with tribal courts. To gauge the representativeness of these findings and to address the possibility that borrowers excluded from the market for small business lending could conceivably substitute to alternative funding sources, we also employ borrower-level data from the FRBNY Consumer Credit Panel. Similarly, consumer credit scores are approximately thirteen points lower (roughly equal to the standard deviation of state-level averages) on reservations under tribal jurisdiction. In addition, data from the FDIC show that community bank branching activity is substantially greater under state courts than under tribal courts. This evidence confirms speculation among lenders (via survey) that more certainty over contract enforcement would improve credit conditions on reservations.[1]

Next, we verify that stronger legal enforcement not only affects credit markets but leads to greater economic activity. Recent research finds that incomes are higher on reservations in which state courts enforce and adjudicate contracts, according to data from the decennial U.S. Census from 1969 to 1999 (Anderson and Parker 2008). We confirm this result using annual, county-level data from the Bureau of Economic Analysis, and an empirical approach that flexibly controls for unobserved regional determinants of economic outcomes by benchmarking the effects of state courts in reservation counties against the economic activity of surrounding counties. Reservation incomes are 10% lower, on average, than incomes in control counties, but state court jurisdiction significantly reduces this gap. Relative to surrounding counties, per capita personal income on reservations under state jurisdiction is 7.1% higher than on reservations under tribal courts. Consistent with the capital market benefits of legal enforcement being particularly important for business activity, proprietor income is more sensitive than overall personal income to court jurisdiction with a differential of 11.2%.

Further, we show the connection between state jurisdiction and reservation incomes works, at least in part, through the effect legal enforcement has on credit market development. If legal enforcement matters for real activity via a credit supply channel, the effects of enforcement should be relatively stronger in the sectors that depend on external capital to fund investment. To evaluate this hypothesis, we build on the insights of Rajan and Zingales (1998) and test whether state court jurisdiction under PL280 has differential effects across industries. Using a variety of proxies for an industry's sensitivity to credit market conditions—including a novel time-varying measure of external finance dependence based on a principal components analysis of industry differences in external finance usage, internal finance generation, and investment intensity—we find that state court jurisdiction disproportionately benefits industries with a greater technological demand for external finance and industries with a larger share of firms facing financing constraints. These cross-sector estimates are robust to reservation-area fixed effects, ruling out a broad class of explanations related to reservation-area unobservables.

Finally, we attempt to quantify the contribution of law-driven finance to economic growth. We use the FFIEC data on small business lending to measure credit market activity at the county level and then rely on differences in court enforcement from PL280 to predict credit market development across reservations. Our empirical estimates from this two-stage approach show that

[1] For example, in a survey of financial services conducted by the Office of the Comptroller of the Currency on Native American Reservations, lenders report that obtaining a better understanding of contract enforcement under the tribal legal system would improve credit conditions on reservations, stating that effective lending requires, "... legal counsel with expertise in Indian law and who can practice in tribal courts" (Native American Working Group 1997).

law-driven improvements to credit markets are associated with significant increases in per capita personal and proprietor incomes. Depending on the estimation approach and sample period, a one-standard-deviation increase in the predicted level of small business credit is associated with increases in personal incomes of 12% to 34%. Moreover, higher levels of predicted credit are associated with differentially higher levels of income in sectors that are more dependent on the supply of external finance. While there are limitations to this approach—namely, there are other potential mechanisms besides credit supply through which court enforcement can affect income—these findings suggest a quantitatively important link between the legal component of credit market development and real economic activity, providing microlevel support for the cross-country evidence in Levine (1998, 1999).

Our paper makes a number of contributions at the intersection of law, finance, and economic growth. Most notably, there is a long-standing interest in understanding the role institutions play in the process of economic development (North 1990; Acemoglu, Johnson, and Robinson 2001; Acemoglu and Johnson 2005). One potential mechanism linking the broad institutional environment with economic performance is the development of the financial sector (King and Levine 1993; Levine and Zervos 1998; Levine 2005), and several prominent studies find that a country's legal and judicial environment affects banking behavior and financial market development (e.g., La Porta et al. 1997, 1998, 2000; Djankov et al. 2002, 2003; Beck, Demirguc-Kunt, and Levine 2003; La Porta, de Silanes, and Shleifer 2006; Haselmann, Pistor, and Vig 2010). However, as La Porta, de Silanes, and Shleifer (2008) discuss, the literature has had more difficulty establishing a causal link between law-driven changes in financial market outcomes and aggregate economic performance. In particular, while several cross-national studies find that the financial market benefits of stronger contract enforcement extend to aggregate economic outcomes (e.g., Levine 1998, 1999; Levine, Loayaza, and Beck 2000), other studies find limited real effects from stronger contracting institutions (Acemoglu and Johnson 2005). By combining detailed area-specific data on credit with plausibly exogenous within-country variation in legal institutions, our work offers novel evidence that the financial market consequences of legal enforcement extend to real outcomes.

This paper also adds to a related literature that evaluates the economic consequences of particular aspects of an economy's legal infrastructure. For example, some recent studies emphasize the importance of stronger legal protections of private property for firm performance and economic growth (e.g., Claessens and Laeven 2003; Berkowitz, Lin, and Ma 2015), and others focus on the benefits of stronger investor protections for real activity at the firm level (e.g., Mclean, Zhang, and Zhao 2012; Brown, Martinsson, and Petersen 2013). Our work turns attention to a less studied aspect of the legal environment: the quality of court enforcement. Research on this topic tends to focus either on broad evidence of court effectiveness in the cross-national context (e.g., Djankov et al. 2003, 2008) or on relatively clean experimental-type evidence on particular effects of within-country shocks to the enforcement environment (e.g., Ponticelli 2013; Gopalan, Mukherjee, and Singh 2014). Our work bridges the gap between these literatures by documenting broad, economically important real effects of court enforcement in a quasi-experimental setting.

Our study adds to an emerging empirical literature that exploits natural experiments and new sources of high-quality data on financial market activity to better understand the determinants and consequences of credit market development (Brown, Stein, and Zafar 2013; Vig 2013; Krishnan, Nandy, and Puri 2015). Our findings on small business credit build on recent insights using home mortgage and consumer credit data on reservations (Parker 2012; Dimitrova-Grajzl et al. 2015), as well as recent work on eligibility for the Community Reinvestment Act and the timing of bank evaluations (Agarwal et al. 2012; Munoz and Butcher 2013), to provide a more comprehensive picture of the robustness of local credit markets under different legal and regulatory environments. A better understanding of the regional determinants of credit market development is particularly important given recent evidence that start-up firms extensively rely on external bank credit (Robb and Robinson 2014) and that better access to bank credit spurs small-firm productivity (Krishnan, Nandy, and Puri 2015). Moreover, by linking the exogenous, law-driven variation in credit market outcomes with long-run economic development, our work speaks to the long-standing interest among financial economists in understanding both the local provision of business credit (e.g., Peterson and Rajan 1994, 1995) and its economic effects (Burgess and Pande 2005; Kerr and Nanda 2009; Butler and Cornaggia 2011; Greenstone and Mas 2012).

Finally, we contribute to an important literature in economics and finance that studies the long-run effects of exogenously imposed differences in geography, culture, and legal rules (Acemoglu, Johnson, and Robinson 2001; Dell 2010; Michalopoulos 2012; Glaeser, Kerr, and Kerr 2015; D'Acunto 2014). Our work is most directly related to the strand of this literature that uses within-country variation to understand the evolution of formal institutions and the cultural and institutional underpinnings of organizational form, firm behavior, and economic performance (Barro and Sala-i-Martin 1992; Bubb 2013; Michalopoulos and Papaioannou 2014; Berkowitz, Hoekstra, and Schoors 2014). Although some of this research also exploits institutional arrangements found on Native American reservations (e.g., Karpoff and Rice 1989; Anderson and Leuck 1992; Cornell and Kalt 2000; Dippel 2014; Cookson 2014), our analysis is among the first to trace out the microlevel mechanisms through which regional differences in institutions matter for both financial and real economic activity. In particular, our work suggests that strong local credit markets can explain much of the broad income differences across reservations with state and tribal courts initially documented by Anderson and Parker (2008), in part by collecting novel, midcentury banking data specific to reservation areas. As such, our findings and approach should be as interesting to policy makers concerned about economic development near reservations, as they are to scholars studying the institutional determinants of cross-national differences in economic performance.

1. Reservation Economies

1.1 Reservation institutions and Public Law 280

Native American reservations are an ideal setting to study the causal effects of institutions. It is appropriate to think of reservations as limited sovereign entities, generally not subject to state laws or regulations, but subordinate to the rule of the U.S. federal government. As a result of a federal policy commitment to tribal sovereignty, the historical status quo is that each reservation runs its own tribal court to enforce the law on that reservation.[2] In addition, reservations are relatively homogenous on unmeasured dimensions due to similar culture and long-term exposure to American institutions, a stark contrast to the extensive heterogeneity in the cross-national setting.

Although reservations have considerable political autonomy, the U.S. Congress passed Public Law 280 in 1953, mandating that a subset of reservations in select states would be subject to jurisdiction by state courts.[3] Legal scholars have suggested that Congress passed PL280 because of a perceived need for stronger criminal enforcement on reservations. According to a 1953 Senate report on PL280: "[… T]he enforcement of law and order among the Indians in Indian Country has been left largely to the Indian groups themselves. In many States, tribes are not adequately organized to perform that function; consequently, there has been created a hiatus in law enforcement authority that could best be remedied by conferring criminal jurisdiction on the States indicating a willingness to accept such responsibility" (Anderson and Parker 2016, 5). As an afterthought to extending criminal jurisdiction, state courts were also granted jurisdiction over civil contract enforcement, "because it comported with the pro-assimilationist drift of federal policy and because it was convenient and cheap [to add to the law]" (Goldberg-Ambrose 1997, 50).

Why did Public Law 280 extend to some reservations and not to others? PL280 was mandated in six states: California, Minnesota, Nebraska, Oregon, Wisconsin, and Alaska (on statehood). In addition, PL280 gave state governments the option to assert PL280 authority after the 1953 law, allowing state courts to hear disputes on reservations within their borders. Between 1953 and 1968, ten states asserted optional PL280 jurisdiction of one form or another, but most of these opt-in assertions of PL280 jurisdiction were limited in scope,

[2] A series of three Supreme Court cases decided by the Marshall Court, called the Marshall Trilogy (between the years 1823 and 1832), formalized this relationship between the U.S. federal government, states, and tribes. Congress has used the authority from the Marshall Trilogy to justify policy interventions on Native American reservations.

[3] The law technically allowed for concurrent jurisdiction between state courts and tribal courts, but, in effect, the introduction of state courts to reservations replaced tribal court activity on PL280 reservations.

for example, applying to only pollution laws or jurisdiction over highways (Jimenez and Song 1998; Getches et al. 1998; Melton and Gardiner 2006). For our purposes, Florida and Iowa successfully asserted PL280 jurisdiction over contractual enforcement, and thus, we include reservations in these states in our measure of state courts. (Anderson and Parker 2016) note that an important reason more states did not assume state jurisdiction under PL280 is that pre-existing disclaimers in many states' constitutions (which were established on statehood) explicitly prohibit jurisdiction in reservation areas. Thus, although court assignment under PL280 was by no means random, the ultimate geographic pattern of PL280 reservations arose in large part from a series of historical accidents.[4]

In all cases in which state courts were granted authority on reservations under PL280, the authority was granted to state courts without tribal consent. In 1968, Congress passed the Indian Civil Rights Act, which contained a provision that required states obtain tribal approval before any additional assertions of PL280 authority. As tribes were unwilling to give up sovereign control of their court systems, there were no additional assertions of state court authority.[5] Consequently, PL280 has resulted in persistent differences in reservation institutions that were not chosen by the tribes themselves.

To maintain the broadest possible sample for our empirical tests, we classify a reservation as under tribal courts if state courts cannot hear civil disputes on the reservation either because the reservation's state never asserted court jurisdiction over native lands or because PL280 jurisdiction was exempted or retroceded as outlined in the 1953 law or in the 1968 amendments to the law in the Indian Civil Rights Act. Otherwise, a reservation is considered to fall under state court jurisdiction. Although our results are robust to alternative categorizations of the law, our main approach is consistent with other studies that have used variation in PL280 civil jurisdiction to study economic outcomes (Anderson and Parker 2008; Cookson 2010, 2014; Parker 2012).[6]

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[4] Beyond limited scope assertions of PL280, both Montana and North Dakota attempted to assert optional PL280 authority. This did not come into force because it conflicted with their state constitutions. In separate legislation (Public Law 785 in 1950), New York reservations were subjected to the state court system. Because we want our measure to reflect whether state versus tribal courts have jurisdiction, we include New York reservations under our measure of state court jurisdiction, but exclude reservations in Montana and North Dakota. In addition, several reservations were exempted from the original law or had court authority retroceded to them.

[5] The 1968 Indian Civil Rights Act also allowed for retrocession of PL280 authority, but the process for retrocession of state court authority to tribal courts is difficult to initiate by tribes. Thus, there were few instances in which tribal court authority was regained. We account for retrocession in our main measure, as well as robustness to alternatives in Table A.8 in the Online Appendix.

[6] Our findings are robust to a number of reasonable alternative classifications, which we show in the Online Appendix. Specifically, we consider classifications that (1) only use variation across mandatory PL280 states and tribal courts, (2) drop retrocessions and exemptions from the sample, and (3) drop observations in Washington that have some form of state jurisdiction, but for which the assertion is less clear (Johnson and Paschal 1992; Anderson and Parker 2016). Throughout our analysis, we employ Cookson's (2010) preferred measure, which retains the broadest sample. For a detailed discussion of important trade-offs in selecting the appropriate classification of state jurisdiction, see Anderson and Parker (2008) or Cookson (2010).

1.2 Law, credit, and economic activity on reservations

Although PL280's contract enforcement implications are not why the law was proposed or passed, the introduction of better-understood state courts to reservations has done much to overcome the unease of investors who are considering signing debt contracts on reservations (e.g., see Anderson and Parker 2008). Within the reservation context, observers have long speculated that problems with credit markets may be attributable to the nature of contract enforcement on some reservations. There is also an impression that improvements to credit markets could improve economic performance. Mudd (1972) describes the likely impacts of two Supreme Court cases involving legal jurisdiction and credit for Montana tribes and portrays the Native American credit problem in the following way: "As a practical matter, non-Indian lenders who face the possibility of using tribal courts to enforce their contracts can be expected to be hesitant in extending credit. The same is true with Indian lenders who in some cases have an equal reluctance to use tribal court. [...] Another view is that the present loss of credit, whether created by the confusion as to where jurisdiction lies, or by lenders' reluctance to rely on tribal courts, is an unfortunate blow to Indians' efforts in economic development and should be remedied."

Moreover, the problem of insufficient credit on reservations persists to this day, with modern policy makers identifying a similar set of challenges (i.e., insufficient legal infrastructure and inability to pledge tribal land as collateral). For example, at a 2010 Senate hearing on the question of Native American unemployment on reservations, the Deputy Assistant Secretary for Indian Affairs Donald Laverdure reported that "The Department of the Treasury conducted a series of workshops, surveys and roundtables to examine Indian access to capital and financial services. Twenty-four percent of American Indians interviewed told the government that business loans were "impossible" to obtain. Treasury's report estimated that the "investment gap" between American Indian economies and the U.S. overall totaled $44 billion. The report also found that, despite the fact that 85% of financial institutions on or near Indian lands offer deposit accounts to American Indian residents, half of those institutions provide only ATMs and personal consumer loans."

The issue of credit on Native American reservations is important unto itself, but, as we argue, a better understanding of the role of credit markets in supporting economic activity on reservations is also more broadly informative about the linkages between law, finance, and growth. In this way, our study of the causes and consequences of credit market outcomes on reservations can speak to settings in which it is much more difficult to measure the causal effects of law and finance.

1.3 Economic conditions by court type before PL280

Although the historical narrative suggests that assignment to state courts under PL280 was unrelated to credit markets and development, it is important to

Table 1
Reservation economies prior to Public Law 280's 1953 passage

Credit and incomes	State courts	Tribal courts	Difference	*p*-value
Bank branches per capita (×1000)	0.0248	0.0313	−0.0065	*0.579*
Bank loans per capita	201.1	191.8	9.29	*0.909*
Bank assets per capita	614.2	596.7	17.51	*0.942*
Family incomes	5.85	5.81	0.04	*0.887*
Economic conditions and demographics				
Nonwhite population (% pop.)	0.0582	0.132	−0.074	*0.001****
High school educated (% pop.)	0.108	0.104	0.0047	*0.535*
College educated (% pop.)	0.0245	0.0270	−0.0026	*0.283*
Unemployment rate	0.0596	0.0601	−0.00053	*0.948*
Fraction urban	0.299	0.301	−0.0011	*0.987*
Fraction incarcerated (×100)	0.583	0.666	−0.083	*0.690*
N	27	75		

This table presents statistics from prior to the passage of the 1953 law, Public Law 280, which gave state courts authority to adjudicate contracts on a subset of Native American reservations. All observations are at the county level. We classify a county as state (tribal) court if Public Law 280 applies (does not apply) to the reservation that has a headquarters in the county. All data come from the 1950 U.S. Census, except for bank branches, bank loans, and bank assets, which come from the 1952 edition of Polk's Bank Directory. The data from Polk's is a county-level aggregate of loans, assets, or branches for banks that are headquartered in that county. These variables are converted to per capita using the county's population according to the 1950 Census. The family incomes measure is the county's median income expressed in terms of income buckets running from zero (lowest income range) to nine (highest). *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

verify that initial conditions on reservations with state and tribal courts were not different in ways that might confound any inferences regarding PL280's long-run impact. Table 1 provides evidence that credit markets, development outcomes, and demographics were broadly similar across state and tribal court jurisdictions prior to PL280's passage. Data on outcomes at the reservation-level around 1953 are scarce, so we focus on the characteristics of the counties in which reservations are headquartered, an approach we rely on throughout the paper.

We evaluate credit markets prior to the 1953 law using hand-collected banking data from the 1952 edition of Polk's Bank Directory (Polks). Polk's includes the name of the bank, the location of its headquarters and branches, and the bank's assets and loans. Using county-level measures of banking activity (bank assets, bank loans, and total number of branches for banks headquartered in the county), the first three rows in Table 1 show that state and tribal court jurisdictions had similar levels of credit market development before PL280. Specifically, per capita bank loans were not statistically different under state courts ($201) versus tribal courts ($192). Bank assets per resident were also similar across jurisdiction ($614 in state and $597 in tribal court counties), as were the number of bank branches per capita in 1952 (0.248 per thousand under state versus 0.313 per thousand under tribal court counties).[7]

[7] In Table A.3 in the Online Appendix we regress county-level banking and income measures on an indicator for whether the reservation was assigned to state courts under PL280. These results also illustrate that initial

Several other potentially relevant characteristics were also similar across jurisdiction, including initial levels of economic development and human capital. Notably, median family incomes in 1949 were almost identical across state and tribal court reservation areas. Additionally, we find similar levels of educational attainment across jurisdiction as measured by the fraction of population with a high school education (10.8% versus 10.4%) or college education (2.5% versus 2.7%). Labor market conditions were also similar, with unemployment rates of 6% in both state and tribal court counties, a figure greater than the 1950 national average of 5.3%. Finally, the shares of urban and incarcerated populations were nearly identical across jurisdiction.

The only notable difference in Table 1 is the population share of nonwhite individuals (13% of the population under tribal courts versus 6% under state courts). On further investigation, this difference arises entirely from our approach to mapping data to reservations (using counties as a unit of measurement to overcome the scarcity of reservation-level data, described in Section 2). Specifically, there is no difference in the nonwhite share across state and tribal jurisdiction for Census tracts that are primarily on reservation land (see Appendix Table A.1 in the Online Appendix, which uses GIS and information from the 2000 Census). Present day incomes under state and tribal courts in Census tracts that do not contain reservation land, but are nonetheless in the reservation's county, are the same. Consequently, including land area beyond the reservation boundary likely attenuates differences between state court and tribal court jurisdictions when measured at the county level. Although the nonwhite share difference in 1950 is concerning at first glance, this evidence gives us confidence that comparing long-run outcomes under state and tribal courts is useful for causal inference.

2. Data and Measurement

2.1 Using county data to study reservation outcomes

We map reservations to U.S. counties because there are neither detailed sector-level data for economic outcomes on reservations nor good measures of business credit available at the reservation level (e.g., see Todd 2012). To link the county-level data to the reservation, we match the location of the reservation's headquarters to its county, using Tiller's Guide to Indian Country (Tiller 1996). The reservation's headquarters is useful for this purpose, because most economic activity on the reservation occurs near its headquarters. We then link the county in which the reservation's headquarters is located to counties that are directly adjacent, as well as those counties nearby (within twenty

conditions on state and tribal reservations were similar prior to PL280. Other research arrives at similar conclusions. For instance, the more aggregate evidence in Parker (2012), Table 2 also supports the conclusion that regions targeted by PL280 did not differ dramatically with respect to initial credit market conditions. He finds that total lending from customary (mostly private) lenders in the 1951 to 1952 period was marginally weaker in Bureau of Indian Affairs (BIA) regions that were predominantly assigned state courts under PL280.

miles) using Collard-Wexler (2014). Because they share common geographic attributes and shocks, but do not share the same institutional environment, these nearby and adjacent counties are a natural control group for use in our specifications.

Although most reservations map well to county land area, Figure 1 illustrates two of the most extreme examples of the imprecision of this exercise. In the first example, the Lake Traverse Reservation (South Dakota-North Dakota) has land in seven counties. However, most of the reservation land is in Roberts County, with little land in any of the surrounding six counties (furthermore, almost all of Roberts County is on the reservation). In the second example, the Hoopa Valley Reservation (California) is wholly contained within Humboldt County, but does not represent a large portion of the county's land. Although most reservations in our sample are about the size of a county, we address the imprecision of the reservation-county mapping by controlling for geographic attributes of the reservation or including reservation fixed effects.

Overall, our sample includes 105 reservation counties, 27 of which have state legal jurisdiction and 78 use tribal courts. Figure 2 presents the geography of these reservation counties across the U.S. Reservations under PL280 status are noticeably scattered across regions of the United States.

2.2 Data sources

2.2.1 Credit market data. We employ a variety of different data sets to describe credit market conditions on reservations. We use data from multiple complementary sources because each credit data set that can be mapped to reservation counties has distinct advantages and disadvantages. Also, even when the data has geographic variation, it is often limited in scope to certain types of borrowers or lenders.

Our primary measure of credit provision is the dollar value of lending to small businesses, which is provided by the Federal Financial Institutions Examination Council (FFIEC) in accordance with the Community Reinvestment Act (CRA). These data are aggregated at the county level and includes loans issued to businesses with less than $1 million in annual revenues, originating from banks that exceed a specified asset threshold.[8] Greenstone and Mas (2012) compare the FFIEC data with FDIC call reports and show that the FFIEC data capture approximately 86% of small business loans.

One shortcoming of the FFIEC's small business data is that, in the presence of credit frictions, business owners could seek alternative sources of funding that are outside the FFIEC's regulatory reach. As such, we supplement the analysis with consumer credit data from the FRBNY Consumer Credit Panel (FRBNY-CCP). This longitudinal data set tracks household liabilities and repayment using a 5% randomized sample of individuals with a social security number and

[8] The asset threshold was $250 million before 2005. In 2007 the threshold rose to $1.033 billion and has since been adjusted annually using the CPI.

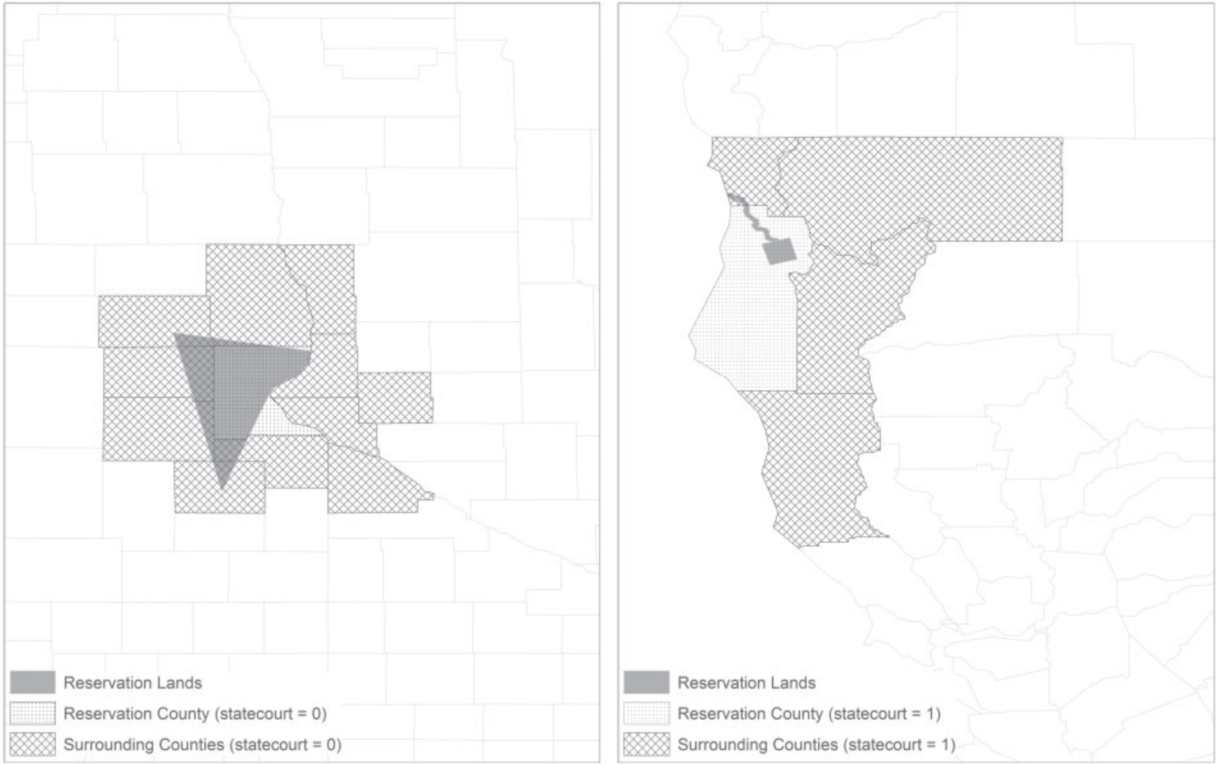

Figure 1
Reservation and surrounding counties
This figure portrays our crosswalk between reservation headquarters and counties, as well as graphically illustrates the empirical strategy employed throughout the regression analysis. In both figures, the darkly shaded region is the reservation's lands according to the TIGER/Line American Indian / Alaska Native / Native Hawaiian Area Shapefiles (AIANNH) in GIS flat file. The county hosting the reservation's headquarters is the reservation county (*resvn*=1), whereas nearby counties are nearby or adjacent counties (*resvn*=0). To illustrate the empirical design, each county in the Lake Traverse Reservation region is labeled "tribal court jurisdiction" (*statecourt*=0, first panel), but only the county in which the reservation's headquarters is located is labeled *resvn*=1. In the second image, every county in the Hoopa Valley Reservation region is labeled "state court jurisdiction" (*statecourt*=1, second panel), and the lightly shaded reservation headquarters county is also labeled *resvn*=1.

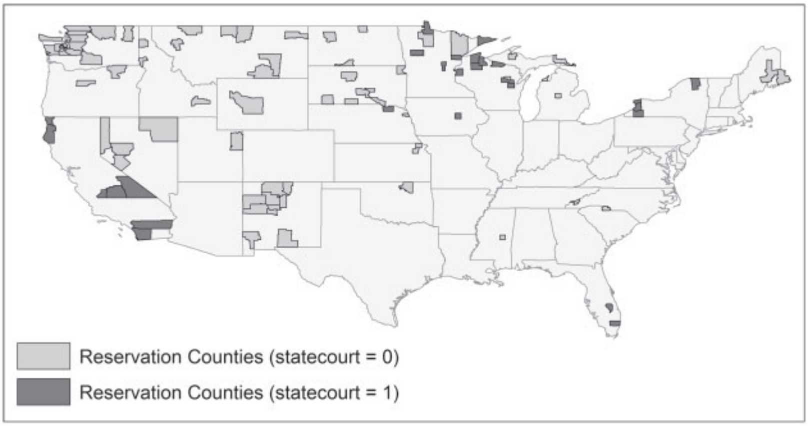

Figure 2
Reservation counties across the United States
This figure shows the distribution of Native American reservations across the United States. The darkly shaded areas are counties in which a reservation under state court jurisdiction is headquartered (*statecourt* = 1). The lightly shaded areas are counties in which a reservation under state court jurisdiction is headquartered (*statecourt* = 0).

a credit report on file at the Equifax credit reporting agency.[9] The data provide a comprehensive picture of reservation-area consumer credit (Dimitrova-Grajzl et al. 2015). Specifically, we use the average consumer risk score in the county, because it reflects a history of borrowing and repayment, and it is a nationally standardized metric on a scale of 300 to 850.

A second limitation of the FFIEC data is that its scope is confined to the lending activity of large banks. For this reason, we provide complementary tests using information from the FDIC Summary of Deposits on the branching activity of community banks that do not meet the threshold for reporting under the CRA. Our measure is the number of full service and retail bank branches, segmented by the bank's total assets.

Throughout the analysis, we collapse each of these data sources to the average of their 1997 to 2003 county-level values (1999–2003 for the FRBNY-CCP data, which start in 1999). We confine the modern-day credit data to these years to make the results comparable across data sets. However, in the Online Appendix, we show that our findings are robust to using the complete range of these data sets in panel form with yearly fixed effects.

2.2.2 Other data sources. In our analysis of economic activity, we employ data from the Regional Economic Information System (REIS, Table CA05),

9 Technically, the sample is randomized by using five pairs of arbitrarily selected digits at the end of an individual's social security number. Moreover, most of the U.S. population has a credit report, and Brown, Stein, and Zafar (2013) provide a favorable view of comparisons between the Equifax data and other nationally representative surveys.

produced by the Bureau of Economic Analysis (BEA). The data include personal income, earnings, and population by county and BEA sector annually from 1969 to 2000.[10] That these data are local, sector-specific, and annual is ideal for studying the effects of courts and credit on economic activity.

The definition of personal income is broader than that of earnings because it also includes proprietor income and income derived from farming, interest, and dividends, as well as from transfers. Within the earnings component of personal income, the REIS data also break down earnings by BEA sector, an industry measure that closely corresponds to one-digit SIC industries but is more refined in some instances (e.g., retail and wholesale belong to the same one-digit SIC industry but are included in separate BEA sectors).

When analyzing sector-specific measures of income, we focus on sectors for which ample economic activity on reservations and their nearby areas can be observed. For this reason, we restrict attention to sectors that have a median personal income across all sample years and counties of greater than $5,000. The sectors that remain in our sample—manufacturing, transportation, construction, retail, and services—comprise the majority of personal income on reservations. Moreover, significant variation occurs across these BEA sectors in the degree to which financing is important for business operations (e.g., firms in the retail sector use considerably less external finance and generate more internal finance than firms in the manufacturing and services sectors). In our analysis of sector income and dependence on external finance, we explicitly use within-reservation variation in personal earnings across BEA sectors to quantify how the provision of credit and legal enforcement matter for economic activity.

In addition to these main measures, we also use data on other aspects of reservations, such as reservation acreage and headquarters locations, from Anderson and Parker (2008) and Cookson (2010). Together with our main measures of credit and income, Table 2 presents descriptive statistics for all the key variables we use in the empirical tests.

2.3 Empirical framework

Given that economic and credit market conditions were similar in state and tribal court counties prior to PL280, the raw difference in current outcomes between state and tribal courts is informative of court enforcement's effect on credit markets and development. However, an important econometric challenge is that broad economic regions (that include reservations) could experience different development trajectories for reasons spuriously correlated to court assignment under PL280. To account for possible region-specific differences in development unrelated to court enforcement, we also evaluate the effect of

[10] Similar county-sector-year-level data are available from 2001 to the present day, but the industry classification changed from SIC industries to NAICS industries. Moreover, the matching between SIC-defined industries and NAICS-defined industries is imperfect, and we avoid having to implement this SIC-NAICS crosswalk by focusing on the SIC-only sample.

Table 2
Summary statistics

Descriptive statistics for the full sample of counties either hosting or adjacent to a Native American reservation

	Mean	Median	SD	5th %ile	95th %ile	Observation
Outcome variables						
Logged small business lending ($ per capita, 1997-2003 avg.)	5.449	5.472	0.813	4.056	6.602	County
Credit score (reservation average, 1999-2003 avg.)	690.1	694.6	23.17	596.4	741.7	County
Logged income ($ per capita) ... panel (1969-2000)	9.200	9.310	0.625	8.090	10.059	Year × county
... cross-section (2000)	10.03	10.04	0.178	9.722	10.302	County
Logged proprietor income ($ per capita) ... panel (1969-2000)	0.827	0.765	0.395	0.312	1.584	Year× county
... cross-section (2000)	1.212	1.165	0.411	0.644	1.966	County
Explanatory variables						
State jurisdiction indicator (statecourt)	0.231	–	0.422	–	–	Reservation
Reservation headquarters indicator (resvn)	0.190	–	0.393	–	–	County
Control variables						
Logged population ... panel (1969-2000)	9.906	9.804	1.390	7.790	12.423	Year× County
... cross-section (2000)	10.037	9.940	1.477	7.724	12.706	County
Reservation acres (100,000s)	6.096	1.997	9.128	0.008	22.874	Reservation
Reservation land in > 2 counties indicator	0.427	–	0.495	–	–	Reservation
Observation counts	Total	State courts	Tribal courts			
Observations in panel	17,629	4,076	13,553			
Observations in cross-section	553	128	425			
# of states	26	7	22			
# of reservations	105	27	78			

Descriptive statistics on sector-level outcomes (reservation counties only)

	Mean	Median	SD	5th %ile	95th %ile	Level
Logged sector income ($ per capita) ... panel (1975-2000)	0.575	0.506	0.373	0.082	1.302	Year × county× sector
... cross-section (2000)	0.783	0.749	0.464	0.000	1.593	County× sector
External finance dependence ... panel (1975-2000)	5.673	5.413	4.398	0.427	12.634	Year× county× sector
... cross-section (2000)	10.246	10.781	4.645	5.408	18.094	County× sector
Observation Counts	Total	State courts	Tribal courts			
Observations in panel	13,435	3,440	9,995			
Observations in cross-section	525	135	390			
# of states	26	7	22			
# of reservations	105	27	78			

This table presents summary statistics of variables we analyze throughout the paper. The data come from the following sources: (1) the Federal Financial Institutions Examination Council (FFIEC), (2) the FRBNY Consumer Credit Panel (FRBNY - CCP), (3) the Bureau of Economic Analysis (BEA) Regional Economic Information System (Table CA05), (4) the 1990 and 2000 U.S. Census and Tiller's Guide to Indian Country as coded in Anderson and Parker (2008), and (5) Compustat. The sample includes counties that contain a reservations headquarters and counties within twenty miles according to Collard-Wexler (2014). State Jurisdiction Indicator is equal to one if the reservation in the county uses the state's court system under the authority of Public Law 280 and zero if the reservation uses tribal courts.

state courts on reservation outcomes against the benchmark provided by nearby counties (e.g., Ellison and Glaeser 1997).

Specifically, we estimate PL280's influence on the gap between reservation outcomes and outcomes in the nearby region using the following spatial difference-in-differences model:

$$Y_i = \gamma_s + \beta_1 resvn_i + \beta_2 statecourt_i + \beta_3 (resvn_i \ x \ \ statecourt_i) + \boldsymbol{\gamma X}_i + \varepsilon_i. \quad (1)$$

In Equation (1), Y_i measures credit market or other economic outcomes for each county i within twenty miles of a reservation's headquarters. In addition to our focus on $resvn_i$ and $statecourt_i$ and their interaction, the model includes state fixed effects (γ_s) and a vector ($\boldsymbol{X}_i$) that contains geographic area and county-level controls. All of our main specifications control for the size of the reservation in acres and the reservation county's population, but in more detailed specifications, we also include the number of counties in which the reservation has land, and in our most stringent specifications, reservation area fixed effects.

The pair of maps in Figure 1 help explain the identification approach employed in Equation (1). Hoopa Valley Reservation falls under state court jurisdiction ($statecourt_i = 1$), while Lake Traverse Reservation does not ($statecourt_i = 0$). Surrounding counties within twenty miles of Hoopa Valley's (Lake Traverse) headquarters are also coded as $statecourt_i = 1$ ($statecourt_i = 0$). Meanwhile, $resvn_i$ equals one for counties in which the Hoopa Valley and Lake Traverse's headquarters are located. The counties that surround the reservation headquarters counties are coded $resvn_i = 0$.

To interpret Equation (1), we are primarily interested in the difference-in-differences coefficient β_3, because it measures how state court jurisdiction affects the gap between reservation and off-reservation outcomes. In the context of our example reservations, β_3 measures the size of the gap between Hoopa Valley and its control counties relative to the size of the gap between Lake Traverse and its control counties. The other coefficients have useful interpretations as well: β_1 reflects the gap between reservation outcomes and off-reservation outcomes under the status quo (tribal courts), and β_2 reflects how regions targeted by PL280 differ from regions not targeted by PL280.

3. Do Courts Affect Credit Markets?

3.1 Legal jurisdiction and reservation credit markets

We start by exploring the link between state court jurisdiction and credit markets on reservations. Following the empirical approach discussed above, the following linear model estimates the effect of state legal jurisdiction on credit market development:

$$credit_i = \gamma_s + \beta_1 resvn_i + \beta_2 statecourt_i + \beta_3 (resvn_i \ x \ \ statecourt_i) + \boldsymbol{\gamma X}_i + \varepsilon_i. \quad (2)$$

The dependent variable, $credit_i$, is either the logged average dollar value of small business loans per capita ($\log(bus.credit_i)$) or the average Equifax risk score of consumers ($riskscore_i$) in county i between 1997 and 2003. To focus our analysis on persistent differences in credit outcomes, we aggregate loan values from the FFIEC business credit data and Equifax risk scores to the county i level, and the estimation includes all counties located within a twenty-mile radius of the county in which the reservation's headquarters is located. In the richest specification, the vector $\boldsymbol{X}_i$ contains logged county population, size of the reservation in acres, an indicator for whether the reservation has land in more than two counties, and the interaction between the multiple county indicator and reservation status. As an alternative approach to controlling for geographic and demographic considerations, and unobservable factors more generally, the following results are robust to reservation-area fixed effects (reported in Table A.4 in the Online Appendix). To re-emphasize our interpretation of the model, the coefficient β_3 on the interaction between $statecourt_i$ and $resvn_i$ captures the impact of state courts on the credit market outcomes relative to the credit market activity in nearby counties. Moreover, we estimate the model using ordinary least squares (OLS) with standard errors allowing for clustering by reservation region (the county in which the reservation's headquarters is located and its associated control counties).

Panel A in Table 3 presents evidence on small business lending. Before estimating Equation (2), we report the estimated effect of state courts on $\log(bus.credit_i)$ in subsamples of reservation-headquarters counties and adjacent counties. The coefficient estimate on $statecourt_i$ from the reservation subsample is 0.363 (SE = 0.171), indicating that small business lending is approximately 40% larger in reservations under state courts than in reservations under tribal courts.[11]

Columns 1 through 4 present estimates of the spatial difference-in-differences Equation (2) with $\log(bus.credit_i)$ as the dependent variable. Regardless of whether the specification includes reservation-area controls (population and reservation acreage), state fixed effects, and multicounty controls (an indicator for more than two counties with reservation land and an interaction with $resvn_i$), the difference-in-differences effect of state jurisdiction is large and statistically significant, with an effect size ranging from 0.35 to 0.44 log-points of business credit. These estimates indicate that business credit is 41.1% to 55.3% greater under state courts than under tribal courts, holding constant the comparison to adjacent counties.

The coefficient estimates on the uninteracted reservation and state jurisdiction dummy variables are also reasonable. The coefficient on the $resvn_i$ dummy variable is significantly negative in each specification, indicative of a sizable reservation credit gap: reservations tend to have less small business lending than do adjacent counties. Consistent with the reservation and adjacent sample splits, the coefficient on the $statecourt_i$ dummy variable is small and statistically insignificant, showing that credit market activity is similar in counties adjacent to reservations with state courts compared with counties

[11] To obtain percentage magnitudes, use the formula from Wooldridge (2003): $exp(\beta)$ - 1, when using a logged dependent variable.

Table 3
The effect of legal institutions on credit

A: The extent of small business lending in different legal environments

	Subsamples		Overall sample			
Dep var: log(*bus.credit*$_i$)	reservation	adjacent	(1)	(2)	(3)	(4)
resvn×statecourt	–	–	0.355**	0.440***	0.392**	0.347*
	–	–	(0.171)	(0.180)	(0.181)	(0.180)
resvn	–	–	−0.268***	−0.410***	−0.376***	−0.253**
	–	–	(0.090)	(0.090)	(0.102)	(0.108)
statecourt	0.363**	0.009	0.009	−0.093	0.081	0.060
	(0.171)	(0.116)	(0.116)	(0.125)	(0.160)	(0.036)
Area controls				x	x	x
State FE					x	x
Multicounty controls						x
R^2	0.035	0.000	0.015	0.092	0.342	0.352
N	104	442	546	546	546	546

B: Consumer credit outcomes in different legal environments

	Subsamples		Overall sample			
Dep var: $riskscore_i$	reservation	adjacent	(1)	(2)	(3)	(4)
resvn×statecourt	–	–	13.06**	12.65**	12.76**	12.79**
	–	–	(5.009)	(5.097)	(4.949)	(4.952)
resvn	–	–	−16.66***	−16.24***	−16.71***	−16.70***
	–	–	(3.794)	(3.907)	(3.717)	(3.725)
statecourt	21.22***	8.162**	8.162**	9.391**	1.491	1.857
	(4.575)	(3.927)	(3.928)	(4.551)	(3.410)	(3.412)
Area controls				x	x	x
State FE					x	x
Multicounty controls						x
R^2	0.129	0.025	0.097	0.101	0.502	0.503
N	105	448	553	553	553	553

This table presents OLS estimation results of the following specification:

$$Y_i = \gamma_s + \beta_1 resvn_i + \beta_2 statecourt_i + \beta_3 resvn_i \times statecourt_i + \boldsymbol{\gamma} \mathbf{X}_i + \epsilon_i,$$

where each observation is either the county in which the reservation's headquarters is located ($resvn_i = 1$) or a county within twenty miles of that county ($resvn_i = 0$), and $statecourt_i$ equals one if the reservation is under PL280 state jurisdiction, and zero otherwise. The vector $\mathbf{X}_i$ contains logged county population, the size of the reservation in acres, an indicator for whether the reservation has land in more than two counties, and the interaction between the multiple county indicator and reservation status to flexibly control for the reservation's effect on adjacent geography. State fixed effects are γ_s. In panel A, the dependent variable log(*bus.credit*$_i$), is the logged per capita loans to small businesses (revenues < $1 million) in the county on average for the years 1997 through 2003 using the FFIEC data. The sample in panel A loses one reservation, Penobscot, MN because there is no FFIEC data for this county. In panel B, the dependent variable $riskscore_i$ is the average consumer risk score (a nationally standardized measure of consumer creditworthiness based on a proprietary formula calculated by Equifax) in the county taken from the FRBNY - CCP averaged over 1999 through 2003. Standard errors are clustered by reservation area. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

adjacent to reservations with tribal courts. The latter finding supports our view that the passage of PL280 was not targeted toward broad regions that experienced more rapid credit market development. Together, the results highlight the relative underdevelopment of credit markets on reservations and show that state court jurisdiction significantly reduces this gap.

Panel B uses our county-level measure of consumer credit ($creditscore_i$) as a dependent variable. As in panel A, the coefficient estimate on $statecourt_i$ in the reservation subsample is positive and statistically significant, indicating that consumer credit markets are stronger under state courts. This conclusion survives our more stringent spatial difference-in-differences specifications, as the estimate for β_3 (the coefficient on the difference-in-differences term $resvn_i$ $x\ statecourt_i$) is also positive and statistically significant across specifications. The estimate for β_3 is approximately 13 credit score points in all specifications, regardless of whether we include reservation-area controls (Columns 1 and 2), or add state fixed effects (Columns 3 and 4) and an indicator for the reservation crossing multiple counties (column 4). To interpret the magnitude of the effect, it is approximately 80% of the coefficient size on $resvn_i$ (approximately 17 points), indicating that state courts alleviate a substantial portion of the reservation credit gap. Another way to assess the economic significance is to note that thirteen points on a credit score is approximately a one-standard-deviation increase in state-level average credit scores across the United States.

We note that the $statecourt_i$ coefficient is also positive and significant in the adjacent subsample regression, as well as in Columns 1 and 2, indicating slightly higher consumer credit scores in areas surrounding reservations with state courts. However, the $statecourt_i$ coefficient estimate becomes close to zero when including state fixed effects (Columns 3 and 4), while the reservation credit-gap estimates are completely unchanged.

3.2 Legal jurisdiction and banking activity

As a robustness check of the main measures of credit market activity, we exploit that the FFIEC data also reports lending activity at the bank-county-year level. This feature of the data allows us to exploit within-bank variation and to evaluate how court jurisdiction affects bank-lending decisions at both the extensive and intensive margins. In particular, does tribal court jurisdiction completely constrain the bank from providing small business loans to reservations, or does most of the aggregated effect arise from originating fewer, smaller loans under tribal courts?

We estimate the difference-in-differences specifications with bank-level fixed effects (γ_b):

$$banking_{ib} = \gamma_s + \gamma_b + \beta_1 resvn_i + \beta_2 statecourt_i + \beta_3 (resvn_i\ x\ \ statecourt_i) + \boldsymbol{\gamma} \boldsymbol{X}_i + \varepsilon_{ib}. \tag{3}$$

Each observation in Equation (3) is a bank-county pair for the set of banks observed in the FFIEC data every year from 1997 to 2003. The dependent variable $banking_{ib}$ is either an indicator for whether bank b lends a positive amount to county i or is the logarithm of one plus the average amount of lending (per capita loans to small businesses with revenues less than $1 million) bank b originates to county i between 1997 and 2003. The vector $\boldsymbol{X}_i$ contains logged county population and the size of the reservation in acres.

Panel A of Table 4 presents estimates of Equation (3) showing that the lending decisions of banks are affected along both margins. Notably, the estimates in

Table 4
Robustness to credit market measurement: The effect of legal institutions on banking activity

A: Banking market characteristics state and tribal court areas

Dep var:	Indicator for any lending			log(per capita lending)		
	(1)	(2)	(3)	(4)	(5)	(6)
resvn×statecourt	0.011**	0.009**	0.009**	0.066**	0.055***	0.055***
	(0.005)	(0.005)	(0.004)	(0.026)	(0.021)	(0.021)
resvn	−0.004*	−0.004**	−0.004**	−0.031**	−0.028**	−0.028**
	(0.002)	(0.002)	(0.002)	(0.013)	(0.011)	(0.011)
statecourt	−0.000	−0.000	−0.000	−0.012	0.000	0.000
	(0.003)	(0.003)	(0.003)	(0.021)	(0.012)	(0.012)
Area controls	x	x	x	x	x	x
State fixed effects		x	x		x	x
Bank fixed effects			x			x
R^2	0.026	0.029	0.127	0.017	0.019	0.133
N	198,360	198,360	198,360	198,360	198,360	198,360

B: Bank branching in state and tribal court areas

Sample:	All banks (1)	Assets < $250M (2)	Assets < $100M (3)
resvn×statecourt	0.232***	0.132*	0.133*
	(0.083)	(0.076)	(0.076)
resvn	−0.208***	−0.162***	−0.161***
	(0.068)	(0.054)	(0.054)
statecourt	0.030	0.001	0.001
	(0.066)	(0.052)	(0.053)
Area controls	x	x	x
State FE	x	x	x
R^2	0.362	0.198	0.195
N	553	553	553

Panel A presents OLS estimation results of the following specification:

$$banking_{ib} = \gamma_s + \gamma_b + \beta_1 resvn_i + \beta_2 statecourt_i + \beta_3 resvn_i \times statecourt_i + \boldsymbol{\gamma} X_i + \epsilon_i,$$

where each observation is a combination of a bank, b, and a county, i. The county is either the county in which the reservation's headquarters is located ($resvn_i = 1$) or a county within twenty miles of that county ($resvn_i = 0$), and $statecourt_i$ equals one if the reservation is under PL280 state jurisdiction, and zero otherwise. In panel A, the dependent variable in Columns 1–3 is an indicator for there being any lending in county i by bank b. In Columns 4–6, the dependent variable is the logarithm of one plus per capita small business using bank-county level. The data used to construct these dependent variables are collapsed to the annual average over the years 1997 through 2003. Panel B presents the same specification, but with the dependent variable of logarithm of one plus the number of branches per 10,000 residents. State fixed effects are γ_s and bank fixed effects are γ_b. Standard errors are clustered by reservation area. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Columns 1 through 3 show that banks are 1.1 percentage points more likely (average propensity to lend is 1.8 percentage points) to originate loans on reservations under state courts than under tribal courts, after benchmarking against lending activity in nearby regions. Further, the regression estimates in Columns 4 through 6 indicate that the intensive margin matters as well: conditional on lending to the reservation, banks extend approximately 6% more small business loans to reservations under state courts than under tribal courts.[12]

[12] In Table A.7 in the Online Appendix, we estimate these effects with tobit regressions to account for a large number of bank-county loan amounts that are equal to zero (i.e., the typical bank only makes loans to a small fraction of the counties in our sample).

Even for the tests that purely rely on within-bank variation, the primary limitation of the small business credit data analyzed in Equation (2) is that the CRA only requires large banks (> \$250 million in assets for years 1997 through 2003) to report small business lending. This reporting threshold is potentially problematic for our analysis of credit provision if state jurisdiction has stronger effects on the decisions of large banks. In particular, it is possible that reduced lending by large banks to areas under tribal courts is at least partially offset by increased lending activity by local community banks.[13] To mitigate this concern, we now analyze how state jurisdiction affects the branching decisions of smaller community banks.

We supplement our analysis of small business lending by estimating the effect of state jurisdiction on the branching decisions of community banks with the following difference-in-differences specification:

$$\begin{aligned} log(1+\ branches.pop_i) = {} & \gamma_s + \beta_1 resvn_i + \beta_2 statecourt_i \\ & + \beta_3 (resvn_i\ x\ \ statecourt_i) + \boldsymbol{\gamma X}_i + \varepsilon_i. \end{aligned} \tag{4}$$

Using data provided by the FDIC Summary of Deposits, the outcome variable, $branches.pop_i$, is the number of bank branches in county i (averaged across the years 1997–2003) per 10,000 county residents. As in our business credit specifications (Equation (2)), each observation is a county i within twenty miles (and inclusive) of the county in which the reservation's headquarters is located, and the coefficient of interest, β_3, reflects the difference-in-differences effect of state court jurisdiction on the extent of community banking activity.

Panel B of Table 4 presents estimates from several specifications of Equation (4). The results indicate a strong and statistically significant effect of state jurisdiction on branching density, regardless of whether we include all banks (Column 1), restrict the count of bank branches to community banks with less than \$250M in assets (Column 2), or only focus on the smallest community banks (< \$100M in assets). As in the business credit specifications, the main effect on $resvn_i$ is negative, showing that reservations tend to have worse financial development (fewer banks per capita of all types) than their adjacent county regions. Our estimates imply that reservations under tribal courts have approximately 20% fewer branches per capita than their adjacent regions, but the reservations under state courts have similar bank branching density relative to nearby counties. That is, the estimates in panel B suggest that the effect of state jurisdiction completely offsets the gap in reservation banking activity.

13 In unreported regressions, we find that larger banks have a relatively greater presence under state courts relative to tribal courts. This pattern is plausible, especially if local community banks are better equipped than are large banks to overcome imperfect local contracting environments using local knowledge of soft information.

Table 5
The effect of state courts on broad categories of income, 1969-2000

Dep var:	log(personal income per capita)				log(proprietor income per capita)			
	1969 - 2000			2000	1969 - 2000			2000
	Reservation	Adjacent	(1)	(2)	Reservation	Adjacent	(3)	(4)
resvn × statecourt	–	–	0.071***	0.058	–	–	0.112***	0.144**
	–	–	(0.027)	(0.038)	–	–	(0.036)	(0.070)
resvn	–	–	−0.108***	−0.105***	–	–	−0.112***	−0.163***
	–	–	(0.022)	(0.027)	–	–	(0.026)	(0.047)
statecourt	0.107***	−0.025	−0.022	−0.026	0.175**	−0.008	−0.001	−0.063
	(0.038)	(0.030)	(0.028)	(0.028)	(0.079)	(0.037)	(0.037)	(0.075)
State FE	x	x	x	x	x	x	x	x
Year FE	x	x	x	–	x	x	x	–
R^2	0.953	0.930	0.930	0.363	0.558	0.503	0.505	0.362
N	3,293	14,336	17,629	553	3,293	14,336	17,629	553

This table presents OLS estimation results of the following specification:

$$log(inc.percap_{it}) = \gamma_s + \gamma_t + \beta_1 resvn_i + \beta_2 statecourt_i + \beta_3 resvn_i \times statecourt_i + \boldsymbol{\gamma}\mathbf{X}_i + \epsilon_{it},$$

where each observation is either a reservation headquarters county ($resvn_i = 1$), or a county within 20 miles of the reservation headquarters county ($resvn_i = 0$), and $statecourt_i$ equals one if the nearest reservation is under the jurisdiction of state courts (zero otherwise). The vector $\mathbf{X}_i$ contains logged county population and the amount of land in acres of the nearest reservation (both standardized to have mean zero and standard deviation one to preserve the interpretation of the main effects). The dependent variable is the logarithm of per capita personal or proprietor incomes from the BEA Regional Economic Information System tables, which we winsorize at the 99th percentile. The sample is annual from 1969 through 2000. Standard errors are clustered by reservation area. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

The results in Table 4 also imply that our findings from the small business credit data are not driven by composition effects within the banking industry. Credit market outcomes improve across the board under state jurisdiction. In particular, state jurisdiction promotes greater branching activity by smaller community banks while at the same time promoting lending by larger banks that meet the CRA reporting threshold. Apart from providing deeper evidence on the positive link between contract enforcement and credit market development, this set of findings supports our use of the small business credit data to measure credit market outcomes across reservations.

4. Do Law and Finance Affect Economic Activity?

4.1 Legal jurisdiction and overall income levels

Anderson and Parker (2008) provide evidence that reservations under state court jurisdiction have higher income growth from 1969 onward. Table 5 updates their findings and extends the analysis to consider proprietor incomes as well. Using our subsample of reservation counties indicated in the columns labeled *reservation*, we estimate that state courts increase reservation personal incomes by 11%, with a noticeably larger effect on proprietor incomes (approximately 18%). Moreover, the incomes in areas surrounding the reservation are no different under state or tribal courts (columns labeled *adjacent*).

The results withstand our more stringent spatial difference-in-differences empirical strategy.[14] The specification is the same as in earlier tests,

$$log(inc.percap_{it}) = \gamma_s + \gamma_t + \beta_1 resvn_i + \beta_2 statecourt_i + \beta_3(resvn_i \ x \ \ statecourt_i) + \boldsymbol{\gamma} \boldsymbol{X}_i + \varepsilon_{it}, \qquad (5)$$

except the dependent variable is now the logarithm of income per capita, $log(inc.percap_{it})$, and year fixed effects (γ_t) account for the panel nature of the BEA income data (1969 to 2000). We present estimates of Equation (5) both for the full range of BEA data (Columns 1 and 3) and to offer a comparison to the tests in Section 3.1 of PL280 on credit, for the 2000 sample (Columns 2 and 4).

In Column 1, the difference-in-differences effect of state jurisdiction on per capita personal income is statistically significant at the 1% level, clustering the standard errors by reservation region. The magnitudes of the estimates are large as well, implying that state jurisdiction has an effect of 7.1% on per capita personal income. Comparing this effect to the $resvn_i$ coefficient estimate (β_1 = -0.107), state jurisdiction overcomes around 70% of the income gap between reservations and their adjacent counties. Although the estimated difference-in-differences coefficient for personal income is marginally insignificant when we use only observations from the year 2000, the magnitude is similar at 5.8% of per capita personal income.

State legal jurisdiction has larger effects on proprietor incomes (Columns 3 and 4). Specifically, the estimate of β_3 is 11.2% of per capita proprietor income in the full sample, and even greater (14.4%) on the sample confined to year-2000 data, both of which are statistically significant at the 5% level or better. Because proprietor incomes are presumably more directly linked to credit market conditions, these results suggest that credit plays an important role in fostering income growth. We now expand our inquiry into the law-finance-growth channel using a more systematic approach.

4.2 Legal jurisdiction, external financial dependence, and sector income

In this section, we evaluate whether the expansion of credit under state courts has led to better economic conditions. To evaluate this finance-growth link, we employ an identification strategy in the spirit of Rajan and Zingales (1998) that utilizes within-reservation variation in incomes across sectors. If state courts promote long-run economic development through a credit channel, then the increases in income under state courts will be stronger in sectors that rely more on external finance. Our cross-sector tests employ several alternative proxies for an industry's sensitivity to the external credit environment, which we describe below.

14 In Table A.2 in the Online Appendix, we also demonstrate this finding using geographically precise measures based on Census tracts mapped to reservation land. The income differences between state and tribal courts are only present in the Census tracts with the most significant overlap with the reservation.

4.2.1 Measures of sensitivity to financial markets. Our first time-varying measure of industry financial dependence, $extfin_{jt}$, is equal to the median firm's ratio of total external financing (net stock and long-term debt issues) to total assets for the past five years. The measure is constructed using a sample of young (in Compustat for less than 15 years), publicly traded U.S. firms that have nonmissing total assets at any point between 1971 and 2000 (company statement of cash flows on external financing is not widely available until 1971). We compute $extfin_{jt}$ from 1975 to 2000, because $extfin_{jt}$ requires four years of data prior to the measurement year.

We also build a second, arguably more comprehensive measure of financial dependence, which incorporates the industry's internal cash flow (cf_{jt}) and fixed investment intensity ($capx_{jt}$) alongside external finance usage ($extfin_{jt}$). We construct a single index by extracting the first principal component of these variables. The first principal component loads on factors that determine dependence on external finance, with an equation given by

$$external.depend_{jt} = 0.773 extfin_{jt} + 0.533 capx_{jt} - 0.346 cf_{jt}.$$

According to this measure, dependence on external finance is greater when the use of external funds is high, investment intensity is high, and cash flow is low.[15]

A potential concern with these external dependence measures, particularly when applied to the reservation setting, is that they are based solely on the financing activities of large, publicly traded firms. As Rajan and Zingales (1998) note when discussing similar concerns in the cross-national context (see 563–564), because U.S. public firms presumably face the least supply-side financing frictions, their financing activities best reflect the sector's technological demand for external finance. Nonetheless, we also consider other characteristics that should affect how sensitive an industry's performance is to credit market development, yet better map to the typical firm located on and around reservations. Specifically, we focus on the degree to which industries are composed of young and small firms. For several reasons, we expect that young and small firms face more severe financial constraints, suggesting sectors with relatively more young and small firms will most benefit from stronger local credit markets. In addition, a measure that exploits variation in the age and size of public firms across sectors is arguably more robust to differences between public and private firms than measures based on the external financing activities of publicly listed firms.

15 When we calculate the PCA, the first two principal components capture over 90% of variation, and they appear to capture distinct effects. The second principal component appears to indicate a tendency of firms to finance investment internally $internal.invest_{jt} = -0.158 extfin_{jt} + 0.688 capx_{jt} + 0.708 cf_{jt}$. We report results using the second principal component in Table A.12 in the Online Appendix.

Hadlock and Pierce (2010) show that an index based only on firm size and age (the SA index) is a reliable predictor of firm-level financing constraints.[16] Following Hadlock and Pierce (2010), we classify firms with an SA index in the top tercile across all firms in a given year to be financially constrained. The measure, $frac.constrained_{jt}$, is the share of total firms in each industry and year that fall into the constrained category. For robustness, we construct proxies for the importance of financing constraints using the median SA index and the median firm size in each industry and year.

4.2.2 Tests of external finance dependence on sector incomes. Using sector-specific income measures from the BEA from 1975 to 2000, we estimate the effect of state jurisdiction across sectors with the specification:

$$log(sector.inc_{ijt}) = \gamma_i + \gamma_{jt} + \beta_1(statecourt_i \ x \ \ extfin_{jt}) + \varepsilon_{ijt}, \qquad (6)$$

where the dependent variable is per capita income in sector j observed for each reservation county i and year t. We restrict the sample in these tests to reservation counties because we can identify the differential effect of state courts from within-county variation in incomes across sectors. As before, $statecourt_i = 1$ indicates that the reservation in county i is subject to PL280 state jurisdiction. The independent variable, $extfin_{jt}$, is a measure of external finance dependence. The baseline specification includes reservation (γ_i) and sector-year (γ_{jt}) fixed effects. Reservation fixed effects account for any time-invariant reservation characteristics that we explicitly control for in other specifications (i.e., reservation size and indicators for reservation land in multiple counties). A positive estimate for our coefficient of interest, β_1, implies that income differences across sectors with high and low $extfin_{jt}$ are larger under state courts than under tribal courts.

Column 1 of Table 6 reports the results from estimating Equation (6) using our primary measure of external finance dependence, $extfin_{jt}$. The estimated coefficient on the interaction $statecourt_i \ \ x \ extfin_{jt}$ is positive and highly significant ($\beta_1 = 0.040$; SE = 0.011), consistent with state jurisdiction promoting economic activity through credit provision to finance-dependent industries. The economic magnitudes are striking. The estimate of Equation (6) implies that the difference in incomes between a sector with high external finance dependence (one standard deviation above the mean) and a sector with average external finance dependence is 4% larger on reservations with state courts compared with reservations with tribal courts. We quantify the magnitude of this 4% effect by noting that it compares favorably in percentage terms to the effect of state courts on aggregate per capita income (7.1%, see Table 5). That the

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16 Using the coefficients from an ordered logit, Hadlock and Pierce (2010) construct the SA index equal to $SA = -0.737size + 0.043size^2 - 0.040age$. *Size* is the (deflated) log of total assets (in millions of \$), and *age* is the number of years since the firm first appears in Compustat with a nonmissing stock price.

Table 6
The effect of legal institutions on sector income, by external finance dependence

Dep var: log($inc.percap_{ijt}$)	(1)	(2)	(3)	(4)	(5)	(6)
Measures of external finance dependence						
statecourt×extfin (Z)	0.040***		0.026***		0.032***	
	(0.011)		(0.010)		(0.010)	
statecourt×extfin.PCA (Z)		0.031***		0.020**		0.029***
		(0.010)		(0.010)		(0.010)
Measures of financial constraints						
statecourt×frac.constrained (Z)			0.043***	0.047***		
			(0.014)	(0.015)		
statecourt×median.firm.size (Z)					−0.047***	−0.051***
					(0.015)	(0.015)
Sector-year FE	x	x	x	x	x	x
Reservation county FE	x	x	x	x	x	x
R^2	0.639	0.638	0.641	0.641	0.642	0.642
N	13,435	13,435	13,435	13,435	13,435	13,435

This table presents OLS estimation results of the following specification:

$$log(sector.inc_{ijt}) = \gamma_i + \gamma_{jt} + \beta_1 statecourt_i \times extfin_{jt} + \epsilon_{ijt}$$

where the unit of observation is county i, sector j, and year t. The sample is annual from 1975 through 2000, and it includes counties that contain a reservation headquarters (excluding nearby counties as in the aggregate tests). The dependent variable, log($inc.percap_{ijt}$), is the logarithm of one plus per capita sector-level incomes from the BEA Regional Economic Information System tables, which we winsorize at the 99th percentile. The sample includes economic sectors that have a median personal income across all sample years and counties of greater than $5,000 (manufacturing, transportation, construction, retail, and services). The independent variable in the specification, *extfin*, is calculated in the following four ways using data from Compustat. In the regression table, *extfin* is computed by aggregating the ratio of firm-level external finance to total assets over the past five years, and then computing the median of this firm-level measure at the BEA sector level. *Extfin.PCA* is the first principal component of yearly *extfin,* the industry's internal cash flow (cf), and fixed investment intensity (capx). *Frac.constrained* is the share of all firms in each industry and year in the top tercile of the SA index in Hadlock and Pierce (2010). *Median.firm.size* is log(assets) for the median firm in the industry-year. For all versions of *extfin*, the variable is scaled to have a mean of 0 and a standard deviation of 1, indicated by (Z) to ease the interpretation of regression estimates. Reservation county and economic sector-year fixed effects are γ_r and γ_{jt}, respectively. Standard errors are clustered by reservation area. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

cross-sector effects are on the same order of magnitude as the aggregate effects suggests that the financing channel highlighted in the cross-sector analysis is an important factor behind the overall difference in incomes across state and tribal jurisdiction. Furthermore, our PCA-based index of external finance dependence leads to similar findings (Column 2, $\beta_1 = 0.031$; SE = 0.010).

In Columns 3 and 4, we include the interaction between state jurisdiction and the share of firms in an industry that Hadlock and Pierce's (2010) SA index classifies as constrained. In each case, β_1 is positive and statistically significant, indicating that state jurisdiction has a stronger effect on income in sectors that have more constrained firms. The magnitude of the cross-sector effects using the financial constraints measure, $frac.constrained_{jt}$, is even stronger than the corresponding cross-sector effects we observe using external finance dependence. Moreover, in Columns 5 and 6 we use median firm size in the industry in place of fraction constrained. The coefficient estimates on the interaction between state court jurisdiction and firm size are approximately −0.05 and are statistically significant at the 1% level. The estimates show that the effect of state jurisdiction is more pronounced in sectors in which the typical firm is smaller. Finally, external finance dependence and firm financing constraints have distinct effects on income. Columns 4 and 6 include $statecourt_i$ x $extfin_{jt}$ as an independent variable alongside the interaction between financial constraints and state courts. In addition to retaining statistical significance, the estimates of $statecourt_i$ x $extfin_{jt}$ have magnitudes similar to those in Columns 1 and 2.

Tables A.10, A.11, and A.12 in the Online Appendix report a number of alternative approaches for estimating the within-reservation, across industry effects of state court jurisdiction. In each case, the results show that state jurisdiction has a positive and relatively stronger effect on incomes in the sectors that *should* be most sensitive to the availability of credit. These cross-sector patterns rule out most other potential mechanisms through which stronger enforcement under PL280 leads to stronger income growth.

4.3 Predicted credit and overall income

We now attempt to quantify the importance of the post-PL280 credit expansion for economic development on reservations. To do so, we project our measures of income onto the law-driven component of credit using a two-stage least squares procedure that exploits the exogenous variation in court enforcement arising from PL280. To the extent that credit expansion is the primary mechanism linking stronger courts with economic development, as our cross-sector tests suggest, this IV-approach provides useful estimates of the quantitative importance of credit expansion for long-run incomes. Although we are reluctant to draw causal inferences from this exercise because other nonfinancial channels correlated with law-driven financing cannot be ruled out, the exercise has the potential to bring us closer to revealing the depths of credit's impact on economic activity.

Table 7 estimates the effect of reservation-area credit market activity on aggregate and sector-specific reservation incomes. The regression specifications and data samples are nearly identical to Equations (5) and (6), respectively. When we use OLS, we replace the independent variable $statecourt_i$ with $log(resvn.credit_i)$, the log of the average dollar value of small business loans per capita for loans made in the reservation headquarters county between 1997 and 2003. Likewise, the IV estimates use $statecourt_i$ to instrument for $log(resvn.credit_i)$ in a two-stage least squares procedure. We normalize $log(resvn.credit_i)$ so that a one-unit increase is equal to a standard-deviation increase in credit.

Panel A presents the results from estimating a specification that mirrors Equation (5).[17] In Column 1, the OLS estimate of the coefficient on $log(resvn.credit_i)$ is 0.122 (SE = 0.037). In Column 2, the IV estimate is 0.341

[17] Because credit shows up in two terms in our spatial difference-in-differences regression specifications, we technically have two endogenous regressors ($log(resvn.credit_i)$ and $resvn_i$ x $log(resvn.credit_i)$). As is standard

Table 7
Quantifying the effect of credit on income

A: Personal and proprietor incomes, 1969-2000

Dep var:	log(personal income per capita)		log(proprietor income per capita)	
	OLS (1)	IV (2)	OLS (3)	IV (4)
resvn×$log(resvn.credit)$ (Z)	0.122***	0.341***	0.184**	0.458***
	(0.037)	(0.042)	(0.033)	(0.068)
resvn	−0.067***	−0.025***	−0.048***	0.006
	(0.015)	(0.008)	(0.017)	(0.013)
$log(resvn.credit)$ (Z)	0.010	−0.50***	0.025	−0.001
	(0.012)	(0.016)	(0.014)	(0.026)
State FE	x	x	x	x
Year FE	x	x	x	x
R^2	0.931	0.924	0.514	0.492
N	17,405	17,405	17,405	17,405

B: Sector-level tests, 1975-2000

Dep var: $log(inc.percap_{ijt})$	OLS (1)	IV (2)	OLS (3)	IV (4)
extfin (Z)×$log(resvn.credit)$ (Z)	0.025***	0.100**	0.025***	0.100***
	(0.010)	(0.014)	(0.010)	(0.012)
extfin (Z)	0.067**	0.068**	0.067**	0.068**
	(0.006)	(0.005)	(0.006)	(0.005)
$log(resvn.credit)$ (Z)	0.066***	0.060***	–	–
	(0.019)	(0.002)	–	–
Year FE	x	x	x	x
Sector FE	x	x	x	x
Reservation county FE			x	x
R^2	0.491	0.446	0.612	0.567
N	13,305	13,305	13,305	13,305

Panels A and B present OLS and instrumental variables (IV) estimates of the regression specifications outlined in Tables 5 and 6, respectively. The level of observation and sample period are the same as in these tables. The regression variables are also the same, except for $log(resvn.credit_i)$, which is the average of small business loans per capita (data from FFIEC) for the reservation county, standardized to have a mean of zero and a standard deviation of 1 (indicated by Z). We use 2SLS to implement the IV regressions, using the variable $statecourt_i$ as an exogenous predictor of $log(resvn.credit_i)$ and $resvn_i \times statecourt_i$ as an exogenous predictor of $resvn_i \times log(resvn.credit_i)$. In all IV specifications, the p-value on the rank-order test (Anderson's canonical correlations test) is less than 0.1%, and thus, the first stage relevance of the instruments is satisfied. Standard errors are clustered by reservation area. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

(SE = 0.042). Accordingly, a standard-deviation increase in business credit is associated with a 12% to 34% increase in personal incomes. Columns 3 and 4 estimate the effect of credit on proprietor incomes. As expected, per capita proprietor income is particularly sensitive to the robustness of credit markets. The OLS and IV coefficients are 0.184 (SE = 0.033) and 0.458 (SE = 0.068).

Panel B mirrors the within-reservation tests in Equation (6).[18] Credit is more strongly related to income in sectors with high external finance dependence.

practice, we use $statecourt_i$ and $resvn_i$ x $statecourt_i$ to instrument for $log(resvn.credit_i)$ and $resvn_i$ x $log(resvn.credit_i)$, and perform the estimation using two-stage least squares.

[18] The IV regressions use $statecourt_i$ and $statecourt_i$ x $extfin$ as instruments for the variables $log(resvn.credit_i)$ and $log(resvn.credit_i)$ x $extfin$.

In regressions with personal incomes as the dependent variable (Columns 1 and 2), the OLS coefficient on *extfin x log(resvn.credit$_i$)* is 0.025 (SE = 0.010) and the IV estimate is 0.100 (SE = 0.014). The coefficient estimates are similar when we also include reservation county fixed effects (Columns 3 and 4). Broadly, the insensitivity of our coefficient magnitudes to richer fixed effects suggests that omitted variable bias is not likely a problem (e.g., see Oster 2014). Moreover, these specifications rule out a wide variety of alternative credit-related mechanisms (i.e., those that plausibly vary at the reservation level, such as the ability to make credible commitments). Because these estimates rely on externally imposed variation in legal enforcement, control for reservation-specific unobservables with reservation fixed effects, and exploit differences across industries in exposure to credit, these findings strongly support the conjecture that legal enforcement promotes economic growth through a finance mechanism.

5. Conclusion

In 1953, the U.S. Congress assigned state courts to enforce contract disputes on a subset of Native American reservations. Given that Native American reservations within the United States are relatively similar along other institutional and cultural dimensions, this setting offers a unique opportunity to evaluate how externally imposed differences in legal institutions affect credit markets and economic activity. Using detailed sector- and location-specific data on credit and income, we document a strong, persistent link between court enforcement and the extent of financial and economic development on reservations. Indeed, the economic magnitude of the effect of court enforcement (7% of per capita income) is remarkable given the overall strength of the U.S. financial sector and the long-standing academic notions that finance finds opportunity and weak contracting institutions have only limited economic effects (e.g., Acemoglu and Johnson 2005).

Much more can be learned from the differential assignment of legal institutions across Native American reservations. For example, we link the strength of court enforcement and development of local credit markets with income levels in finance-dependent sectors because this analysis is a logical starting point for understanding how law and finance broadly promote growth. In a similar vein, future work might extend our approach to study how access to credit affects new firm creation, entrepreneurial activity, employment, and productivity growth, particularly in finance-dependent sectors. In addition, given small-firm reliance on bank credit (e.g., Robb and Robinson 2014) and the importance of small business enterprises for reservation-area employment and income, the reservation setting is well suited to evaluate how household-level collateral constraints and financial health influence the creation and growth of new enterprises (e.g., Hurst and Lusardi 2004; Adelino, Schoar, and Severino 2015).

We exploit variation that arises from sharp historical differences between state courts and tribal courts on reservations, but this paper's lessons undoubtedly apply broadly. Namely, our work suggests that improved court effectiveness can be a particularly important facilitator of growth in settings in which legal institutions are relatively weak. Moreover, the quantitative importance of the effects we document suggests that the courts may continue to influence economic performance even in settings, such as across U.S. states, in which the institutional variation is less pronounced. For example, state legal systems differ with respect to how much judicial independence is afforded to the appointment and retention of judges (Hanssen 2004), and these differences lead to notable variation in court outcomes (e.g., tort awards studied in Tabarrok and Helland 1999). Appealing to this paper's findings, these differences may also have first-order effects on the real economy.

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